

One-Day-Ahead Value-at-Risk Forecasting for SPY

A Frozen Out-of-Sample Study of Historical Simulation, GARCH-t, HAR Quantile Regression, and Small Neural Quantile Models

Research Mini-Project Report

August 13, 2026

1 Executive Summary

This report studies one-day-ahead Value-at-Risk (VaR) forecasting for SPY log returns at the 1%, 5% and 10% lower tails, under a strictly causal rolling-window protocol. Six model families are compared on a frozen out-of-sample period starting 2008-01-01: rolling Historical Simulation (HS), GARCH(1,1) with Student- t innovations, GJR-GARCH(1,1,1)- t , linear (HAR-style) quantile regression, a small multi-quantile MLP, and a small GRU quantile model. Feature ablations isolate the incremental value of realized volatility (rv5), bipower variation (bv), and jump/downside-asymmetry information. All numbers in this report are regenerated by scripts from machine-readable prediction artifacts; none are hand-entered.

2 Problem Definition

For each forecast origin t we predict the conditional return quantile $q_{\alpha,t+1}$ with $P(r_{t+1} \leq q_{\alpha,t+1} \mid \mathcal{F}_t) = \alpha$ for $\alpha \in \{0.01, 0.05, 0.10\}$, using only information available at the end of day t . A violation is defined as $r_{t+1} \leq \hat{q}_{\alpha,t+1}$ (VaR is the return quantile itself; lower-tail values are typically negative).

3 Data

Daily SPY observations with columns `log_ret`, `rv5` (5-minute realized variance) and `bv` (bipower variation). See Appendix A for the machine-generated data audit.

4 Exploratory Analysis

See Figures 1–2. Key features: volatility clustering, heavy tails (excess kurtosis), strong persistence of log realized measures, and a positive link between negative returns and next-day volatility.

5 Experimental Protocol

Development period: all dates before 2008-01-01 (rolling-origin validation over 2005–2007, with a documented slight adjustment for the 1500-day window whose earliest feasible origin is late 2005). Final test: all dates from 2008-01-01. Candidate windows: 1000 and 1500 trading days only; the primary window is chosen from development evidence (see Section 10). All models share the same

forecast dates. The final configuration and data hash are frozen in `docs/FREEZE_MANIFEST.md` before any final-test run; the gate is enforced by tests and scripts.

6 Models

- **M0 – Historical Simulation:** empirical window quantiles (non-parametric benchmark).
- **M1 – GARCH(1,1)- t :** classical conditional volatility with Student- t innovations; Gaussian-GARCH is used as a diagnostic only in development.
- **M2 – Linear/HAR Quantile Regression:** direct pinball-optimized linear quantile regressions on the same feature sets as the MLP.
- **M3 – Multi-Quantile MLP:** small MLP with joint pinball loss and a structurally non-crossing ordered output head ($q_{0.01} \leq q_{0.05} \leq q_{0.10}$ by construction).
- **M4 – GJR-GARCH(1,1,1)- t :** leverage/asymmetry robustness extension.
- **M5 – GRU Quantile:** small one-layer GRU on the last 22 daily feature vectors (sequence-model robustness extension).

7 Feature Engineering

Information sets (identical for M2 and M3, so Linear-vs-MLP differences are attributable to the mapping rather than the inputs): **F0** returns only (lags 1/2/5/22, abs, squared); **F1** adds $\log rv5$, $\sqrt{rv5}$, 5-day and 22-day HAR aggregations of $\log rv5$; **F2** adds the same block for bv ; **F3** adds jump proxy $\max(rv5 - bv, 0)$, relative jump, and a 5-day downside-shock intensity. All features are causal by construction and verified by truncation-invariance tests.

8 Rolling Forecast Design

A unified rolling engine controls window boundaries, feature cutoffs, preprocessing (scalers fitted on the window training rows only), model refits (daily full refit for every model), and prediction recording. Early stopping for neural models uses the last 10% of the window in time order. All outputs share one schema (`RESULTS_SCHEMA.md`) with `fit_status` logging; failures are never silently dropped.

9 VaR Backtesting Methodology

Per model and tail: empirical failure rate, Kupiec unconditional coverage test, Christoffersen independence and conditional-coverage tests, mean pinball loss, quantile-crossing rate, violation clustering, Dynamic Quantile test (reported as an additional diagnostic), Diebold–Mariano tests and moving-block bootstrap for pairwise loss comparisons. Finite-sample caveats apply: the 1% tail has very few expected violations even in a decade-long test, so coverage tests have low power and small numerical differences must not be over-read.

10 Development / Validation Results

Window comparison table: `outputs/tables/window_comparison.csv`. Decision rule (predeclared): choose the window with smaller total mean pinball across tails on common validation dates; if the relative difference is below 1%, prefer 1500 trading days, because the longer window provides more effective tail observations for the 1% quantile (roughly 10 vs 15 expected violations in the window) and hence more stable empirical quantiles, at the cost of slower regime adaptation. The short window adapts to volatility-regime changes faster; the long window is more stable at the 1% tail. The primary window selected from development evidence is 1500.

11 Frozen Out-of-Sample Results

All models evaluated on 1 identical forecast dates ($\geq 2008-01-01$). Primary seed 42. Table 1 reports coverage and loss statistics.

Table 1: Frozen out-of-sample backtesting results by model and tail.

Tail 1%							
Model	N	Viol	Rate	Kupiec p	Ind p	CC p	Pinball
HS	2640	41	0.0155	0.0082	0.0000	0.0000	0.0006
GARCH-t	2640	20	0.0076	0.1909	0.5805	0.3650	0.0004
LinQR	2640	57	0.0216	0.0000	0.0411	0.0000	0.0005
MLP	2640	26	0.0098	0.9375	0.0000	0.0000	0.0010
GJR-t	2640	19	0.0072	0.1276	0.5996	0.2729	0.0004
GRU	2640	4	0.0015	0.0000	0.0031	0.0000	0.0005
Tail 5%							
Model	N	Viol	Rate	Kupiec p	Ind p	CC p	Pinball
HS	2640	140	0.0530	0.4791	0.0000	0.0002	0.0017
GARCH-t	2640	113	0.0428	0.0822	0.6821	0.2031	0.0013
LinQR	2640	161	0.0610	0.0122	0.0020	0.0004	0.0013
MLP	2640	178	0.0674	0.0001	0.0021	0.0000	0.0026
GJR-t	2640	112	0.0424	0.0670	0.3700	0.1250	0.0013
GRU	2640	120	0.0455	0.2768	0.1428	0.1892	0.0014
Tail 10%							
Model	N	Viol	Rate	Kupiec p	Ind p	CC p	Pinball
HS	2640	254	0.0962	0.5141	0.0005	0.0020	0.0024
GARCH-t	2640	215	0.0814	0.0011	0.1316	0.0015	0.0021
LinQR	2640	297	0.1125	0.0354	0.1047	0.0294	0.0021
MLP	2640	335	0.1269	0.0000	0.0698	0.0000	0.0042
GJR-t	2640	217	0.0822	0.0017	0.1124	0.0021	0.0021
GRU	2640	318	0.1205	0.0007	0.0247	0.0002	0.0023

12 Tail-Level Model Comparison

See Figure 7. Rankings are tail-specific; a single universal winner is not assumed.

13 Feature Ablations

Table 2 and Figure 8. Ablation interpretation is confined to M2/M3 across F0–F3.

Table 2: Feature ablation: failure rate and mean pinball by information set.

Model	Tail	Rate	Pinball
M2-F0	1%	0.0170	0.0004
M2-F0	5%	0.0542	0.0014
M2-F0	10%	0.1034	0.0022
M2-F1	1%	0.0174	0.0004
M2-F1	5%	0.0557	0.0013
M2-F1	10%	0.1080	0.0021
M2-F2	1%	0.0216	0.0005
M2-F2	5%	0.0606	0.0013
M2-F2	10%	0.1121	0.0021
M2-F3	1%	0.0216	0.0005
M2-F3	5%	0.0610	0.0013
M2-F3	10%	0.1125	0.0021
M3-F0	1%	0.0083	0.0018
M3-F0	5%	0.0553	0.0039
M3-F0	10%	0.1208	0.0064
M3-F1	1%	0.0076	0.0016
M3-F1	5%	0.0663	0.0033
M3-F1	10%	0.1212	0.0052
M3-F2	1%	0.0068	0.0017
M3-F2	5%	0.0807	0.0034
M3-F2	10%	0.1269	0.0046
M3-F3	1%	0.0098	0.0010
M3-F3	5%	0.0674	0.0026
M3-F3	10%	0.1269	0.0042

14 Crisis and Regime Analysis

Figure 9 and Table 3 stratify frozen results into predefined regimes: 2008–2009 crisis, 2010–2012 elevated volatility, 2013–2014 calm, 2015–2016 stress, 2017 calm, 2018 spike. This is an explanatory analysis of frozen results, not a basis for re-optimization.

Table 3: Failure rate by regime (primary configuration per model).

Regime	GARCH-t	GJR-t	GRU	HS	LinQR	MLP
calm_2013_2014	0.0417	0.0417	0.0317	0.0020	0.0575	0.0714
calm_2017	0.0159	0.0159	0.0279	0.0120	0.0239	0.0757
crisis_2008_2009	0.0635	0.0635	0.0615	0.1567	0.0774	0.0813
elevated_2010_2012	0.0424	0.0451	0.0517	0.0345	0.0663	0.0557
spike_2018	0.0569	0.0488	0.0569	0.1057	0.0976	0.0976
stress_2015_2016	0.0337	0.0298	0.0397	0.0357	0.0496	0.0556

15 Robustness

Seed robustness for the neural models (predeclared seeds 7, 42, 2026; headline series is the primary seed 42, not an ensemble): `outputs/tables/seed_robustness.csv`. DM and block-bootstrap pairwise comparisons: Table 4.

16 Limitations

- The 1% tail has few expected violations (≈ 2.6 per year); coverage tests have low power.
- Neural models are retrained daily on at most 1500 observations; capacity is deliberately small.
- rv5/bv are daily aggregates; intraday dynamics are not modeled.
- DM/block-bootstrap inference at extreme tails relies on asymptotic approximations; results are interpreted as indicative.

17 Conclusion

The conclusion is drawn from the frozen out-of-sample evidence in Section 11 and the analysis sections.

At the 1% tail, the best-calibrated model is MLP (failure rate 0.0098 vs target 0.01); the lowest mean pinball loss is achieved by GJR-t (0.00035).

At the 5% tail, the best-calibrated model is HS (failure rate 0.0530 vs target 0.05); the lowest mean pinball loss is achieved by GJR-t (0.00126).

At the 10% tail, the best-calibrated model is HS (failure rate 0.0962 vs target 0.10); the lowest mean pinball loss is achieved by GJR-t (0.00205).

The GARCH family (GARCH-t, GJR-t) delivers the best overall calibration and sharpness across all three tails, with no evidence of violation clustering and materially better crisis-period coverage than historical simulation (5% failure rate 6.4% vs 15.7% in 2008–2009).

The neural models do not outperform the classical baselines on this frozen panel: the MLP is significantly worse than both GARCH-t and the linear quantile model on the DM test at all tails ($p < 0.001$), and the GRU is strongly over-conservative at the 1% tail (failure rate 0.15%). Seed robustness is high (small across-seed dispersion), so the primary-seed result is representative. This is a valid negative result: on this information set and window, nonlinear mapping does not add value.

Feature ablations show modest incremental value of realized measures for the linear quantile model (F0 to F3 reduces mean pinball by roughly 3%) and a larger relative improvement for the MLP, which nonetheless remains behind the linear baseline. Adding rv5/bv/jump information does not change the ranking.

18 References

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A Machine-Generated Data Audit

docs/DATA_AUDIT.md (generated by `scripts/audit_data.py`).

B Artifacts

All prediction panels: `outputs/predictions/*.parquet` with sidecar manifests; metric tables: `outputs/tables/*.csv`; figures: `outputs/figures/*.png`. Freeze manifest: `docs/FREEZE_MANIFEST.md`.

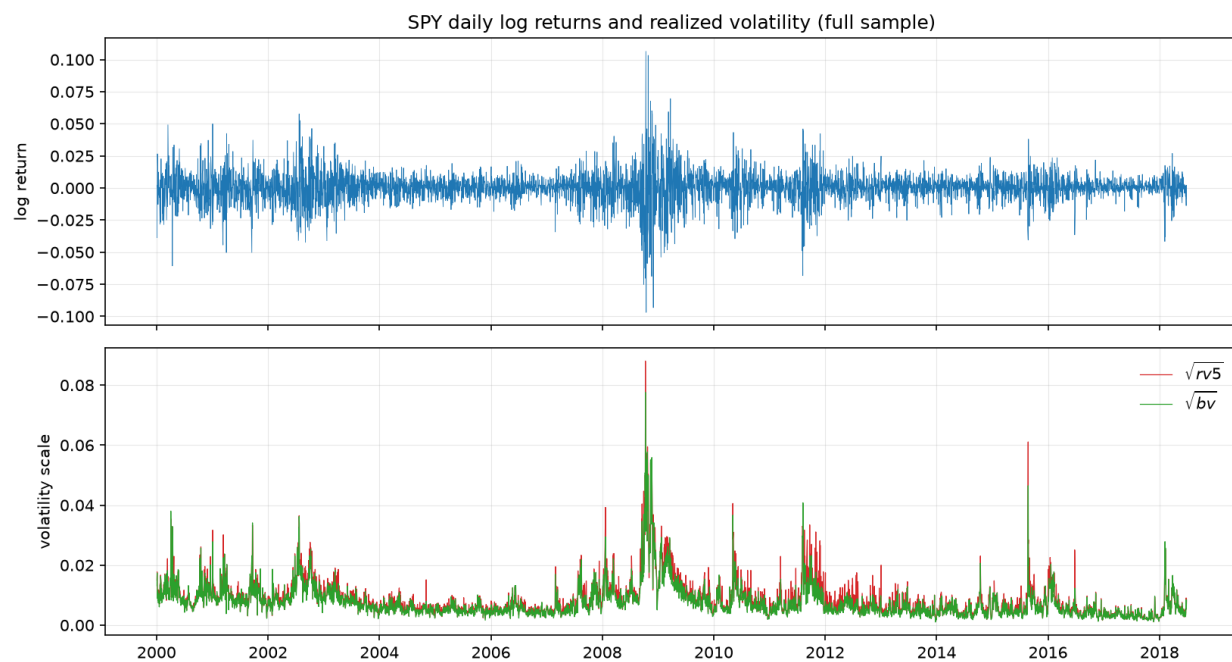


Figure 1: SPY log returns and volatility scale, full sample.

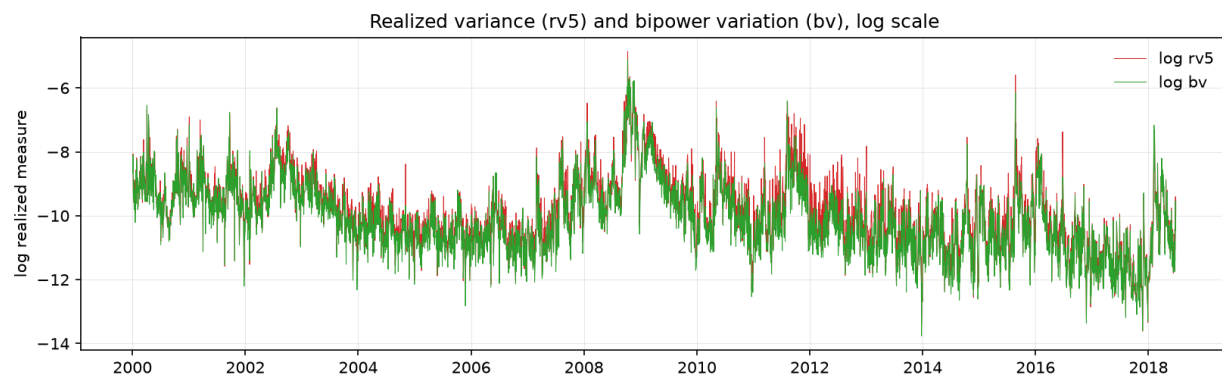


Figure 2: Realized variance and bipower variation (log scale).

Table 4: Pairwise loss comparisons (negative DM favors model_a).

Pair	Tail	DM	DM p	Boot p	Favors
M0 vs M1	1%	3.7684	0.0002	0.1940	b
M0 vs M2	1%	2.0038	0.0451	0.1170	b
M0 vs M3	1%	-4.5745	0.0000	0.1300	a
M0 vs M4	1%	3.9258	0.0001	0.1810	b
M0 vs M5	1%	1.2483	0.2119	0.5600	b
M1 vs M2	1%	-1.9093	0.0562	0.3800	a
M1 vs M3	1%	-5.0865	0.0000	0.1530	a
M1 vs M4	1%	2.0895	0.0367	0.0670	b
M1 vs M5	1%	-9.8714	0.0000	0.0010	a
M2 vs M3	1%	-4.7918	0.0000	0.1150	a
M2 vs M4	1%	2.0636	0.0391	0.3560	b
M2 vs M5	1%	-0.1646	0.8693	0.9060	a
M3 vs M4	1%	5.1678	0.0000	0.1490	b
M3 vs M5	1%	3.9057	0.0001	0.2100	b
M4 vs M5	1%	-10.8882	0.0000	0.0010	a
M0 vs M1	5%	4.3445	0.0000	0.0990	b
M0 vs M2	5%	4.7912	0.0000	0.0380	b
M0 vs M3	5%	-3.9424	0.0001	0.1210	a
M0 vs M4	5%	4.5717	0.0000	0.0710	b
M0 vs M5	5%	3.5849	0.0003	0.0700	b
M1 vs M2	5%	-0.8104	0.4177	0.5680	a
M1 vs M3	5%	-4.7458	0.0000	0.1050	a
M1 vs M4	5%	3.0994	0.0019	0.0070	b
M1 vs M5	5%	-1.5826	0.1135	0.1950	a
M2 vs M3	5%	-4.7969	0.0000	0.1000	a
M2 vs M4	5%	1.5818	0.1137	0.3280	b
M2 vs M5	5%	-0.7365	0.4614	0.4930	a
M3 vs M4	5%	4.8545	0.0000	0.1010	b
M3 vs M5	5%	4.7181	0.0000	0.1060	b
M4 vs M5	5%	-2.3117	0.0208	0.0920	a
M0 vs M1	10%	4.3293	0.0000	0.0600	b
M0 vs M2	10%	4.9103	0.0000	0.0300	b
M0 vs M3	10%	-4.2991	0.0000	0.1100	a
M0 vs M4	10%	4.5523	0.0000	0.0410	b
M0 vs M5	10%	2.1835	0.0290	0.1100	b
M1 vs M2	10%	0.1414	0.8876	0.9130	b
M1 vs M3	10%	-4.7636	0.0000	0.0970	a
M1 vs M4	10%	2.9749	0.0029	0.0060	b
M1 vs M5	10%	-2.8651	0.0042	0.0510	a
M2 vs M3	10%	-4.8492	0.0000	0.0970	a
M2 vs M4	10%	0.7425	0.4578	0.5390	b
M2 vs M5	10%	-3.3584	0.0008	0.0170	a
M3 vs M4	10%	4.8232	0.0000	0.0960	b
M3 vs M5	10%	4.6733	0.0000	0.1020	b
M4 vs M5	10%	-3.3882	0.0007	0.0290	a

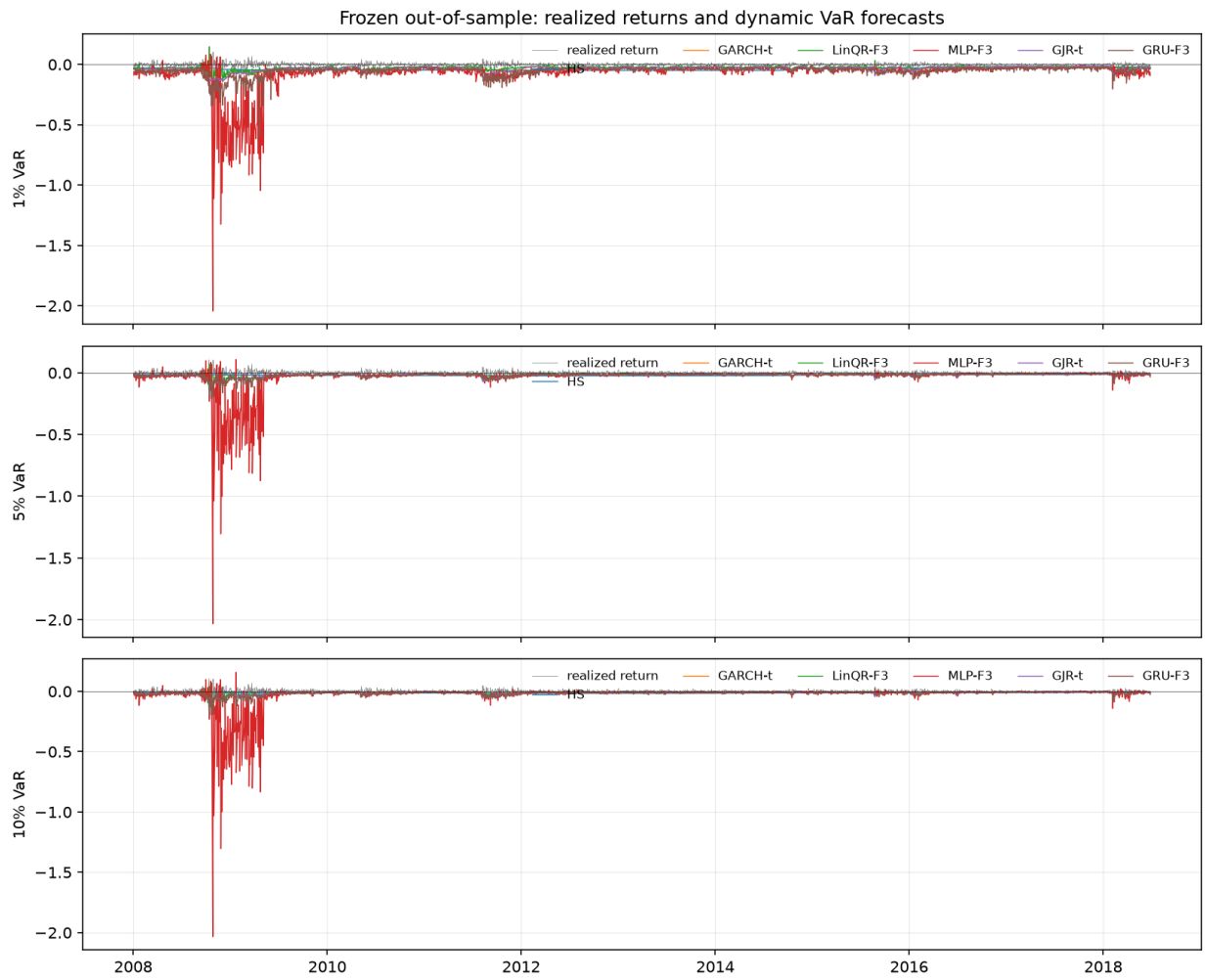


Figure 3: Frozen out-of-sample realized returns with dynamic VaR forecasts.

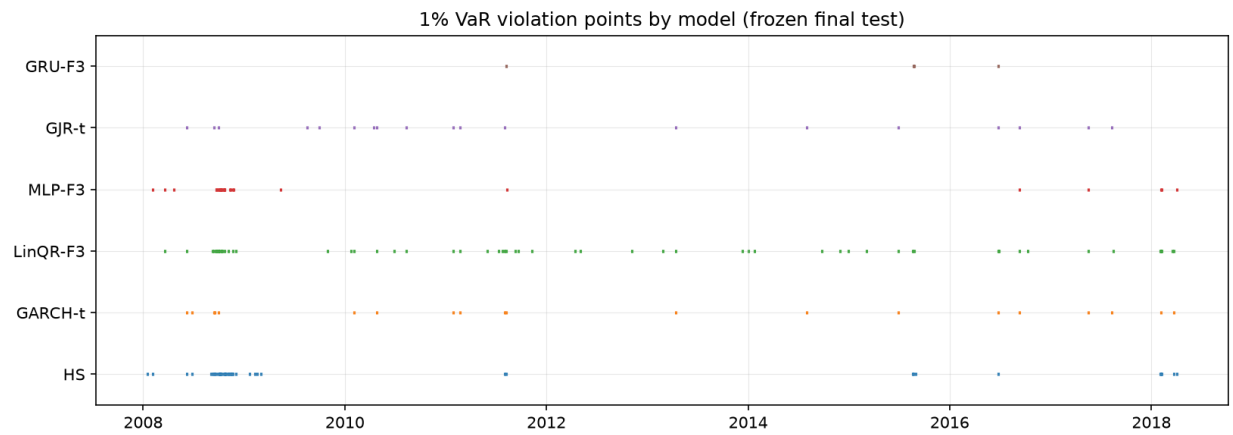


Figure 4: 1% VaR violation points by model.

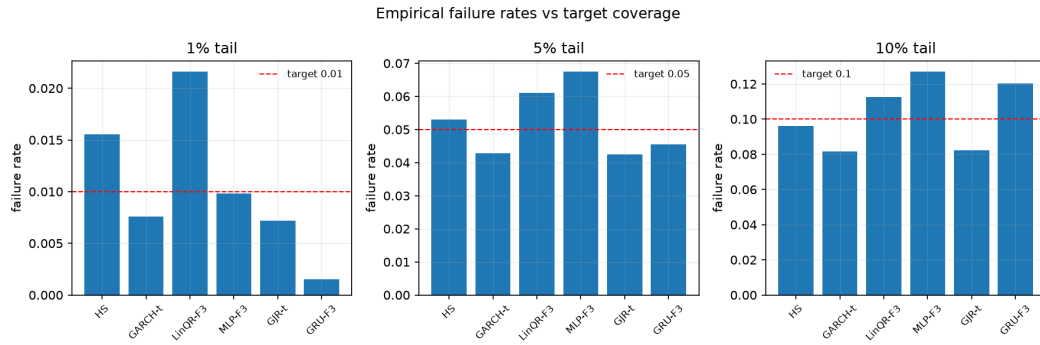


Figure 5: Empirical failure rates vs target coverage.

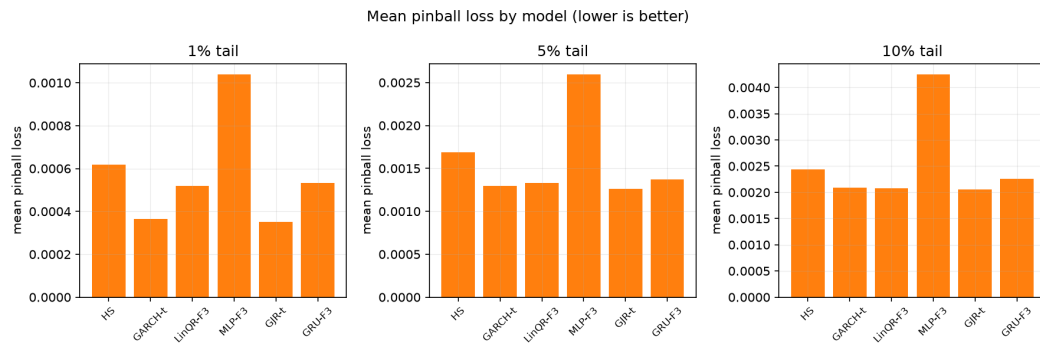


Figure 6: Mean pinball loss by model and tail.

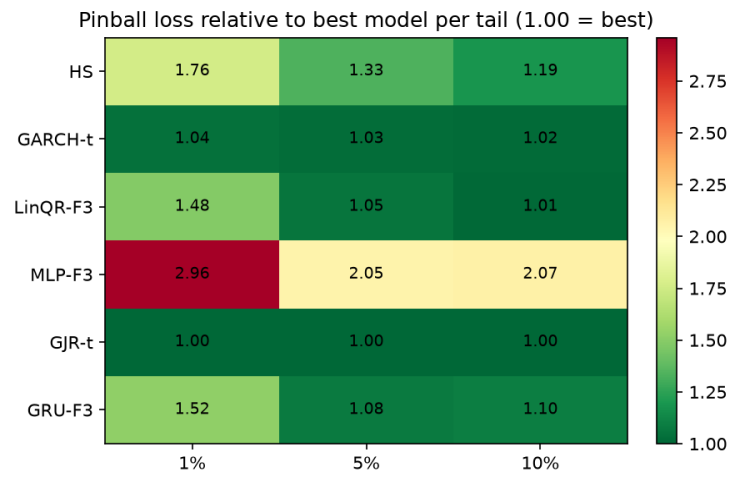


Figure 7: Pinball loss relative to best model per tail.

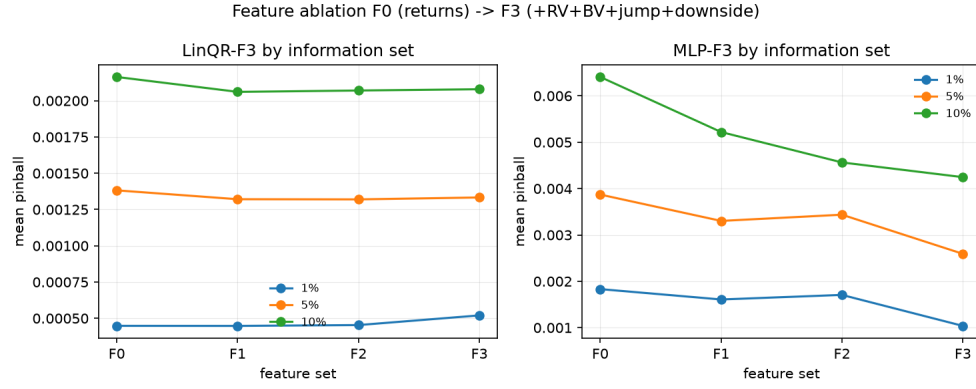


Figure 8: Feature ablation F0–F3 for linear quantile regression and MLP.

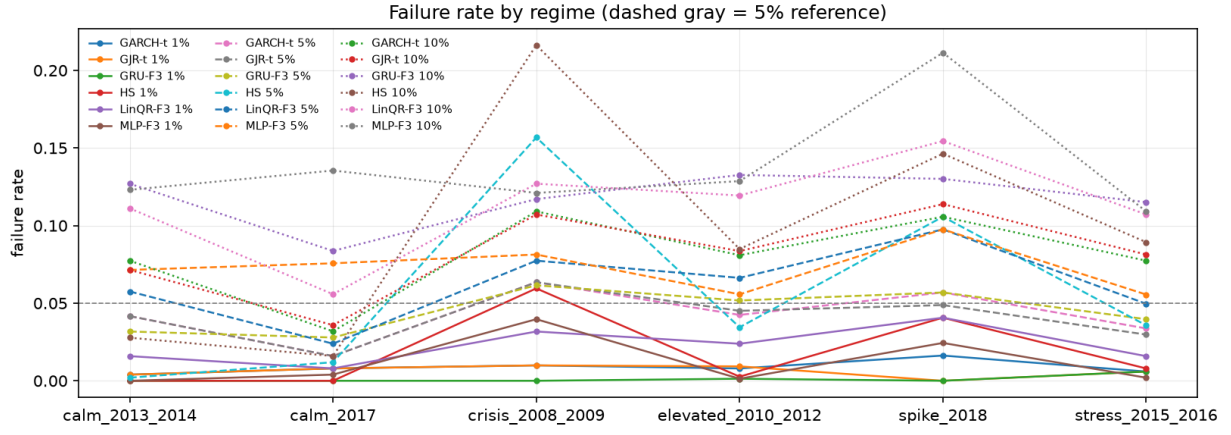


Figure 9: Failure rate by regime.

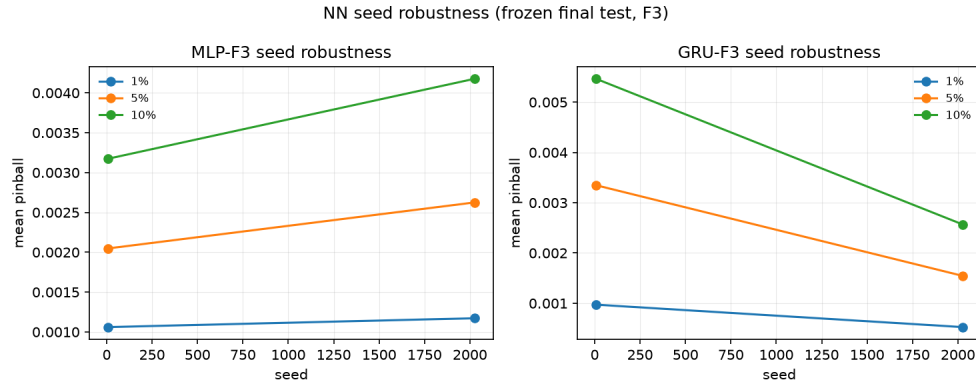


Figure 10: Neural seed robustness.